Lab Experiment 6: Implement Undirected Graphical Model

1. Aim:

To understand and implement an **Undirected Graphical Model (UGM)** using Markov Random Fields (MRFs) for a simple image denoising application.

2. Objectives:

- To understand the concept of **Undirected Graphical Models (UGMs)** and their probabilistic structure.
- To implement **Markov Random Fields (MRFs)** for structured probabilistic modeling.
- To apply **MRFs in image denoising** using **Graph Cuts and Gibbs Sampling**.

3. CO Mapping:

CO5 – Analyze and apply the undirected graphical model.

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4. Introduction to Undirected Graphical Models (UGMs)

Undirected Graphical Models (UGMs), also known as **Markov Random Fields (MRFs)**, represent **dependencies** between random variables in a **graph-based structure** without any directed edges. Unlike Bayesian Networks, which use **conditional probabilities**, UGMs use **potential functions** or **energy-based representations** to describe relationships.

UGMs are widely used in **computer vision, natural language processing, and probabilistic inference** applications.

Key Characteristics of UGMs:

- **Nodes represent random variables** (e.g., pixel values in an image).
- **Edges represent dependencies** between variables without a directed structure.
- Uses **clique potentials** and **energy functions** to define joint distributions.

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5. Problem Statement:

Given a **noisy grayscale image**, we use a **Markov Random Field (MRF)** model to **denoise** it using **Gibbs Sampling**. Each pixel is a node, and edges represent **spatial dependencies** between neighboring pixels.

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6. Understanding Markov Random Fields (MRFs) for Image Denoising

- A **Markov Random Field (MRF)** assumes that a pixel's intensity depends **only on its neighbors**.
- We model this as a **grid graph**, where each pixel interacts with its **4 nearest neighbors**.
- The **energy function** of the system is given by:

$$E(x) = \sum_{i} \phi(x_i) + \sum_{(i,j)\in E} \psi(x_i, x_j)$$

  where:

  - $\phi(x_i)$ is the **unary potential** (likelihood of pixel value).
  - $\psi(x_i, x_j)$ is the **pairwise potential** (smoothness constraint).
  - The goal is to **minimize the energy** to find the best denoised image.

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7. Implementation

Steps:

1. Load the **noisy grayscale image**.
2. Construct a **Markov Random Field (MRF) graph** where pixels form nodes.
3. Define **energy functions** using unary and pairwise potentials.
4. Use **Gibbs Sampling** for iterative denoising.
5. Display **original and denoised images**.

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8. Python Implementation

10. Conclusion

- **Markov Random Fields (MRFs)** provide a powerful way to model spatial dependencies in images.

- The **Gibbs Sampling technique** allows efficient probabilistic inference for denoising.
- MRFs are widely used in **image processing, computer vision, and structured probabilistic modeling**.

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11. Viva Questions

1. What is the main difference between Directed and Undirected Graphical Models?
2. Why do we use Markov Random Fields (MRFs) in image denoising?
3. How does the Gibbs Sampling process work in MRF-based denoising?
4. What is the role of the **beta parameter** in the pairwise potential function?
5. Can MRFs be used for applications other than image processing? Provide examples.

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12. References

1. **Kevin Murphy**, *Machine Learning: A Probabilistic Perspective*, MIT Press, 2012.
2. **Christopher M. Bishop**, *Pattern Recognition and Machine Learning*, Springer, 2006.
3. **D. Koller & N. Friedman**, *Probabilistic Graphical Models: Principles and Techniques*, MIT Press, 2009.